Neither half of the N=96 cut is load-bearing alone

The two cuts interact, and the corner that fails to train is the one keeping its MLP.

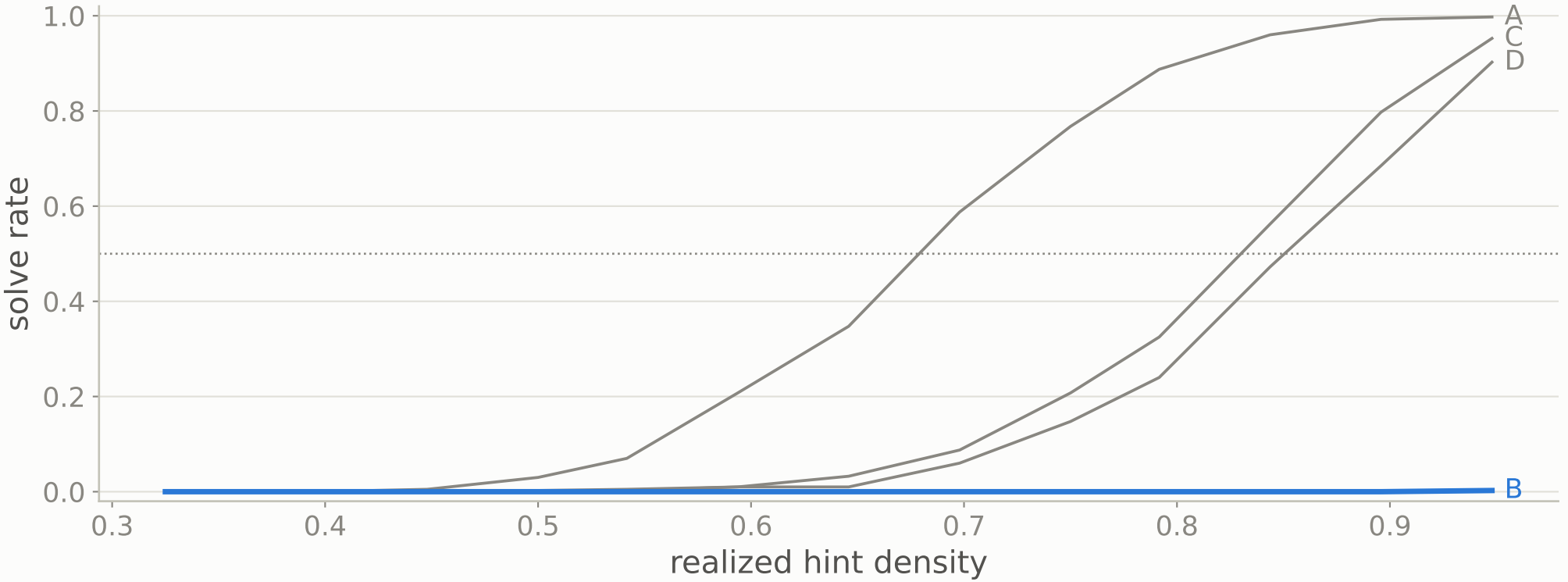
array 51745334 | scorer 51746081 | commit d325907 | 3 seeds/corner | N=96 d5 q4 --shuffle-adj --fall 240k x 50ep | 400/bin

The deep-narrow full model is the only corner that fails

corner	rho_50	crossing	mean_solve	val_acc
A full L12 d256	0.676	3/3	0.4229	0.4822
B full L16 d192	**CENSORED**	**1/3**	**0.0002**	**0.4370**
C no_mlp L12 d256	0.830	3/3	0.2146	0.4721
D no_mlp L16 d192	0.850	3/3	0.1811	0.4709

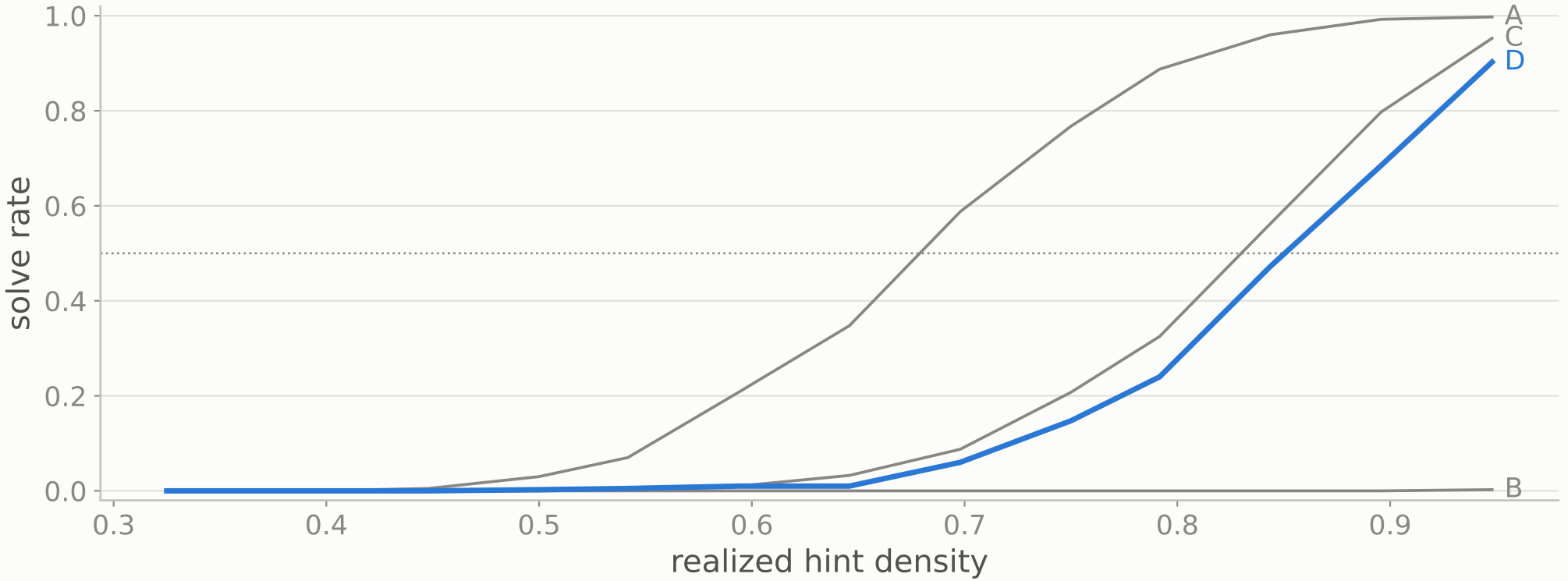
Lower rho_50 is better. B censors on 2 of 3 seeds, so its level is not quotable and M0's main effect does not exist.

Removing the MLP rescues the shape that kills the full model



B (full L=16 d192) never reaches 0.5. D is the same shape with no MLP at all and crosses on every seed.

Attention-only trains at L=16 d192 where the full model censors



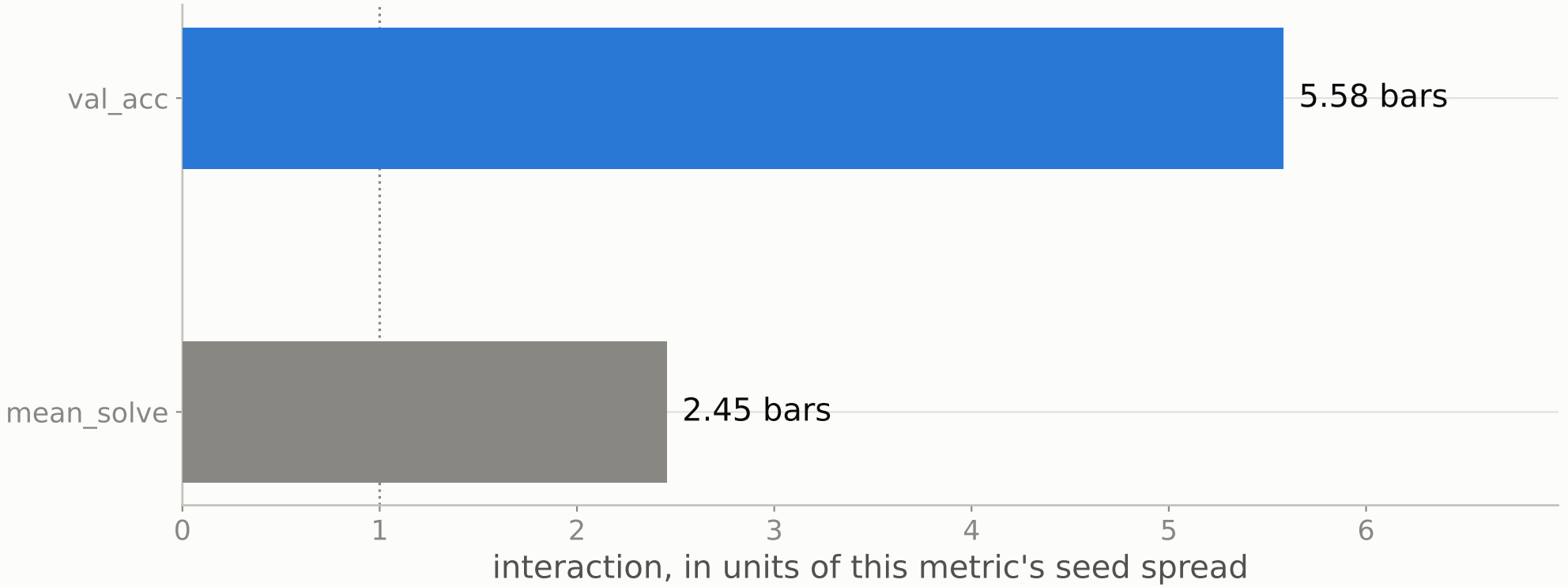
Same depth, same width, same budget, same seeds. The only difference is that D's feedforward writes zero.

The two cuts are sub-additive: the pair costs less than the parts

metric	dMLP (C-A)	dSHAPE (B-A)	sum	actual (D-A)	interaction
rho_50	+0.154	censored	--	+0.175	not readable
mean_solve	-0.2082	-0.4227	-0.6309	-0.2418	-0.3891
val_acc	-0.0101	-0.0452	-0.0554	-0.0113	-0.0441

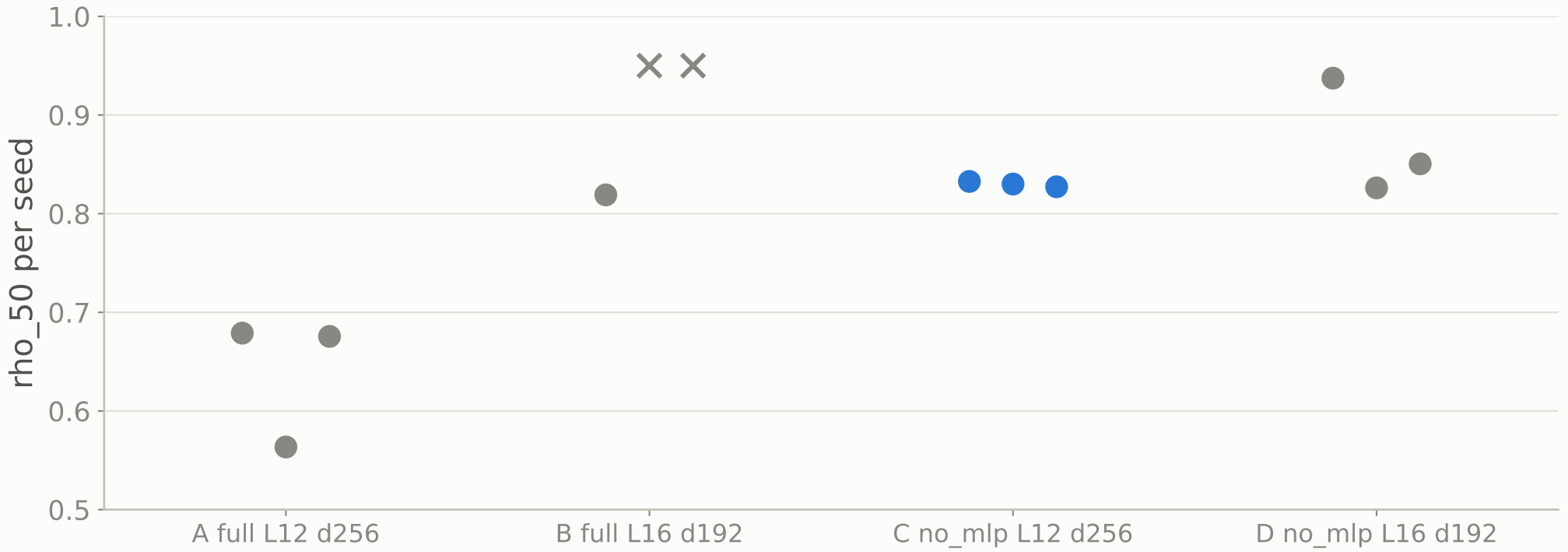
rho_50 has no additivity row because corner B censored. The two never-censoring metrics both read sub-additive.

Interaction reaches 2.45 bars on solve rate, 5.58 on accuracy



Dotted line is one seed spread. An interaction this far past it means no single-factor statement here generalizes.

Corner C's three seeds agree to 0.005, the tightest rung yet

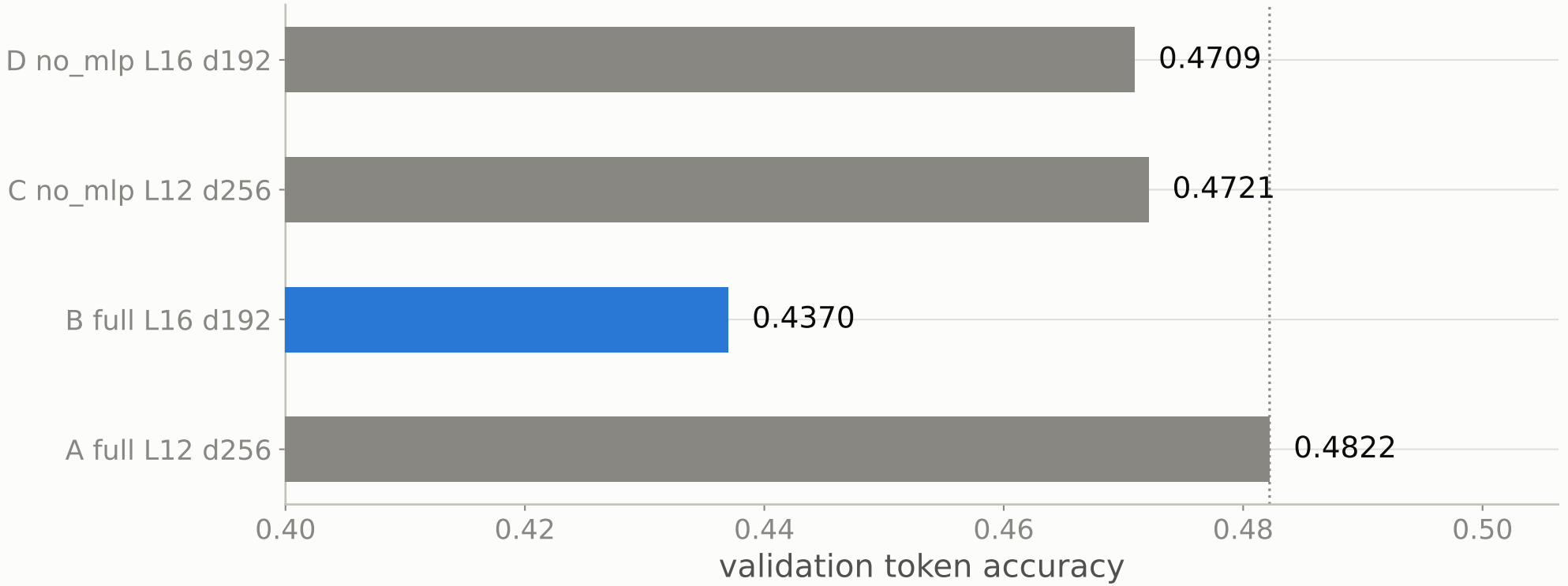


B's two censored seeds are drawn as crosses at the grid top; they have no level, and that is the corner's result.

M0 cannot be scored as registered because corner B censored

- M0 asked that exactly one of |dMLP|, |dSHAPE| clear 0.117
- dSHAPE is not a level: 2 of 3 B seeds never reach 0.5
- Registered outcome (d) binds -- the cost is an interaction

Validation accuracy ranks the corners exactly as solve rate does



Accuracy never censors, so it orders B too. Dotted line is the control. B is worst on every metric, including this one.

What the model actually sees

Every token is one categorical id into a 106-symbol vocabulary (N=96, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	`BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING`
	`BEGIN_ROUND SOLVED UNRESOLVED`
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	`COLOR_0 COLOR_1 COLOR_2 COLOR_3`
	the q colours (token name COLOR_n, prose 'colour')
10-105	`N0 N1 ... N95`
	one id per node, no structure in the id

GRAMMAR

`BEGIN_HINTS`	`(NODE COLOR)*`
`BEGIN_GRAPH`	`(NODE (NODE){degree})*`
`BEGIN_REASONING`	`(BEGIN_ROUND (NODE COLOR)+)*`
`SOLVED`	

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 778 TOKENS

HINTS	`BEGIN_HINTS N0 COLOR_1 N1 COLOR_2 N2 COLOR_2 N7 COLOR_1 N8 COLOR_0 N10 COLOR_3 ... (113 tokens)`
GRAPH	`BEGIN_GRAPH N0 N1 N41 N4 N45 N84 N1 N92 N42 N0 N38 N15 ... (577 tokens)`
REASONING	`BEGIN_REASONING BEGIN_ROUND N3 COLOR_3 N17 COLOR_3 N22 COLOR_3 N24 COLOR_2 N25 COLOR_0 N31 ... (88 tokens)`

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 6 rounds, 40 deductions,
80 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000015, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

56 hints of 96 nodes, then 6 rounds of simultaneous deduction. Hints: N0=COLOR_1 N1=COLOR_2 N2=COLOR_2 N7=COLOR_1 N8=COLOR_0 N10=COLOR_3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND        N3 COLOR_3 N17 COLOR_3 N22 COLOR_3 N24 COLOR_2 ...
  17 deduced
BEGIN_ROUND        N4 COLOR_2 N9 COLOR_3 N11 COLOR_1 N40 COLOR_0 ...
  9 deduced
BEGIN_ROUND        N6 COLOR_1 N14 COLOR_2 N15 COLOR_0 N34 COLOR_2 ...
  5 deduced
BEGIN_ROUND        N5 COLOR_1 N29 COLOR_3 N51 COLOR_2 N75 COLOR_3
  4 deduced
... 2 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N3, neighbours N28 N95 N87 N25 N50

their colours known before this round:

```
   [0, 1, 2]  (3 = q-1 of the 4)

   (4 known neighbours, 3 DISTINCT colours -- some repeat)

   still unknown: N25
```

only colour left: 3

```
so the trace emits  N3 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=96, degree 5, 4 colours, --shuffle-adj --fall: the distribution all four corners were trained and evaluated on.

T=6 rounds, the median of 40 draws from this config (range 1-23) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

Finding 55's gap is an interaction, not the MLP cut

quantity	stage F (previous best read)	stage M (this run)
what was compared	D vs A only	all four corners of the 2x2
rho_50 gap D-A	+0.175	+0.175, unchanged
attributed to	**the attention-only cut**	**an interaction (2.45 bars)**
mean_solve gap	-0.2418	-0.2418, unchanged
MLP cut alone	never measured	+0.154 of that +0.175

Same eval, same grid, same bars. F's numbers do not move; what moves is what they may be attributed to.

The one-cut-at-a-time ladder cannot answer this at N=96

- Depth-16 width-192 needs the MLP removed to train at all
- So 'which half is load-bearing' has no answer at N=96
- Every architecture claim here stays an N=48 claim